



i don't have ground truth labels

beside PCA and UMAP both works for visualization right?

and choose only top pipelines for me to do to test, not all 30, for ex PCA might not so good with hdbscan, umap works better with hdbscan?

i will test for only 6 pipeline i think with only 1 pipeline have kmeans involved, other will test with all other clustering methods, choose which (pca/umap) fit best with each clustering method for dimensional reduction and appropriate for surrogate_models also

Perfect! With **no ground truth labels**, you'll rely entirely on **unsupervised clustering metrics** (Silhouette, Davies-Bouldin, Calinski-Harabasz).

Yes, **both PCA and UMAP work for visualization** as well as for dimensionality reduction before clustering.

Recommended 6 Pipelines (Based on Research)

Based on empirical evidence, here are the **best DR + clustering combinations**:

Pipeline	DR	Clustering	Surrogate	Why This Combo
1	PCA	KMeans	Random Forest	PCA proven to work well with KMeans ^[1] ^[2] ; RF is robust for KMeans labels
2	UMAP	HDBSCAN	XGBoost	UMAP + HDBSCAN = best results ^[3] ^[4] ; XGBoost handles density-based clusters well
3	UMAP	Spectral	XGBoost	Spectral + UMAP = highest performance on complex manifolds ^[3]
4	UMAP	Agglomerative	Random Forest	UMAP improves v-measure for all combinations ^[4] ; RF stable for hierarchical clusters
5	UMAP	GMM	Decision Tree	UMAP preserves structure for probabilistic clustering ^[3] ; DT handles soft boundaries
6	PCA	HDBSCAN	Random Forest	PCA baseline for HDBSCAN (to compare with UMAP-HDBSCAN) ^[5]

Key Insights from Research

UMAP vs PCA by Clustering Method:

Clustering	Better DR	Evidence
HDBSCAN	✔ UMAP	2–3× improvement in ARI with UMAP [3]; UMAP + HDBSCAN = best results [4]
Spectral	✔ UMAP	Spectral + UMAP = highest absolute performance [3]
Agglomerative	✔ UMAP	UMAP increased v-measure for all combinations [4]
GMM	✔ UMAP	UMAP outperforms PCA across all algorithms [3]
KMeans	✘ PCA	PCA + KMeans = classic, proven combo [1] [2]

Overall Pattern:

- UMAP consistently outperforms PCA across all clustering algorithms [3]
- PCA showed surprisingly poor performance in projection-based clustering [6]
- Nonlinear DR (UMAP) > linear (PCA) for complex manifold data [3]

Your 6-Pipeline Testing Matrix

```
pipelines = [  
    {"name": "PCA-KM-RF",    "dr": "PCA",    "cluster": "KMeans",    "surrogate": "RF"},  
    {"name": "UMAP-HD-XG",   "dr": "UMAP",   "cluster": "HDBSCAN",   "surrogate": "XG"},  
    {"name": "UMAP-SP-XG",   "dr": "UMAP",   "cluster": "Spectral",   "surrogate": "XG"},  
    {"name": "UMAP-AG-RF",   "dr": "UMAP",   "cluster": "Agglomerative", "surrogate": "RF"},  
    {"name": "UMAP-GM-DT",   "dr": "UMAP",   "cluster": "GMM",       "surrogate": "DT"},  
    {"name": "PCA-HD-RF",    "dr": "PCA",    "cluster": "HDBSCAN",   "surrogate": "RF"}  
]
```

Evaluation Metrics (No Ground Truth)

Since you have no labels, use these unsupervised metrics:

Metric	Formula	Better	What it measures
Silhouette Score	$(b - a) / \max(a, b)$	Higher (−1 to 1)	Cohesion vs separation
Davies-Bouldin Index	$\max(\text{mean_dist_i} / \text{dist_ij})$	Lower	Intra vs inter-cluster distance
Calinski-Harabasz	$\text{var_between} / \text{var_within} * (n - 1) / k - 1$	Higher	Variance ratio
Inertia	$\sum \text{dist}(\text{point}, \text{centroid})^2$	Lower	Sum of squared distances

Primary metric: Use Silhouette Score as your main comparison metric. [7] [8]

What to Compare

For each of your 6 pipelines, measure:

1. **Silhouette Score** (main metric)
2. **Davies-Bouldin Index** (secondary)
3. **Calinski-Harabasz Index** (secondary)
4. **Runtime** (computation time for each pipeline)
5. **Number of clusters found** (especially for HDBSCAN)

Then create:

- **Bar chart:** Silhouette scores for all 6 pipelines
- **Table:** All metrics side-by-side
- **Key finding:** UMAP vs PCA comparison for HDBSCAN (pipelines 2 vs 6)

Expected Results (Based on Literature)

Based on research, you should expect:

Pipeline	Expected Silhouette	Reason
UMAP-HD-XG	✔ Highest	Best combo for density-based clustering ^[3] ^[4]
UMAP-SP-XG	✔ High	Spectral + UMAP = top performance ^[3]
UMAP-AG-RF	✔ High	UMAP improves all combos ^[4]
UMAP-GM-DT	⚖ Medium-High	UMAP outperforms PCA ^[3]
PCA-KM-RF	⚖ Medium	Classic PCA-KMeans, but PCA underperforms overall ^[6]
PCA-HD-RF	✖ Lowest	PCA poor for HDBSCAN (vs UMAP) ^[3]

Final Recommendation Summary

✔ **Your 6 pipelines are optimal** for comparing without ground truth:

- 1 PCA + KMeans (classic baseline)
- 4 UMAP + different clustering (showing UMAP's dominance)
- 1 PCA + HDBSCAN (for direct UMAP vs PCA comparison)

✔ **Use Silhouette Score** as your primary metric

✔ **Expected finding:** UMAP pipelines will outperform PCA pipelines, especially for HDBSCAN, Spectral, and Agglomerative ^[4] ^[6] ^[3]

This gives you a **comprehensive yet focused comparison** suitable for a formal report!

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1. <https://qiita.com/bmj0114/items/db9145a707cb6ed13201>
2. <https://epubs.siam.org/doi/pdf/10.1137/1.9781611972740.54?download=true>
3. <https://arxiv.org/html/2503.23215v1>
4. <https://ailb-web.ing.unimore.it/icpr/media/slides/11549.pdf>
5. <https://umap-learn.readthedocs.io/en/latest/sources/faq.rst.txt>
6. <https://publikationen.ub.uni-frankfurt.de/opus4/frontdoor/deliver/index/docId/86577/file/1-s2.0-S2352914824001291-main.pdf>
7. <https://www.geeksforgeeks.org/machine-learning/clustering-metrics/>
8. https://medium.com/@Sunil_Kumawat/performance-metrics-for-clustering-9badee0b7db8
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10. <https://medium.com/@ps.deeplearning.training/hdbscan-clustering-and-umap-visualisation-f31e653f7218>
11. <https://arxiv.org/abs/2402.04582>
12. https://adeit-estaticos.econgres.es/20_EAM/Posters/47776_MarinaSanchez.pdf